Практическая работа 3

Сквозная трассировка микросервисов

2.1. Здоровый запрос

Трасса `f22f161`, `GET /api/checkout`, 158,31 мс, HTTP 200.

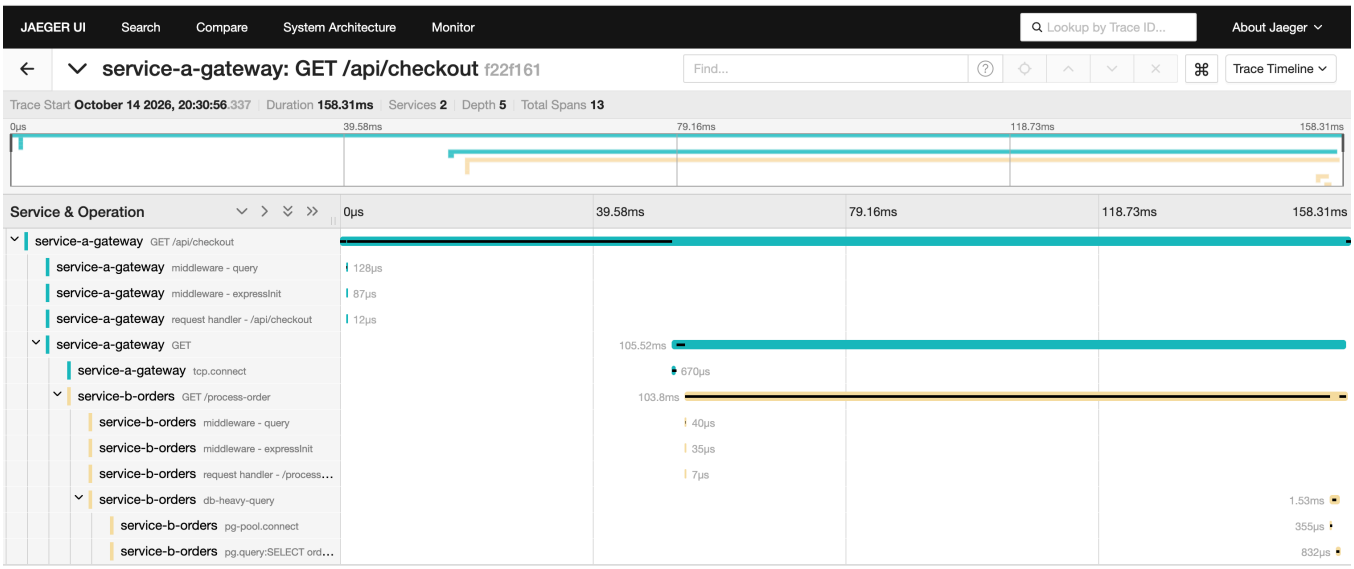


Рис. 1. Трасса `f22f161`.

Вопросы

- Сколько всего спанов содержит трасса? **13**.
- Каким HTTP-заголовком передался контекст? **`traceparent`**. В Jaeger значения нет, поэтому в `service-b` добавлен лог всех заголовков запроса (`console.log(req.headers)`). В логе: `00-36a6fd22152fbc03652f308df54b390a-8d17a26a3c1230f0-01`.
- Сколько занял SQL `SELECT * FROM orders ...`? **832 мкс**.

```
service-b-1  | Service B listening on port 4002
service-b-1  | headers: {
service-b-1  |   traceparent: '00-36a6fd22152fbc03652f308df54b390a-8d17a26a3c1230f0-01',
service-b-1  |   host: 'service-b:4002',
service-b-1  |   connection: 'close'
service-b-1  | }
```

Рис. 2. Лог заголовков входящего запроса в `service-b`.

2.2. Узкое место

Трасса `e24d797`, 2,07 с, HTTP 200.

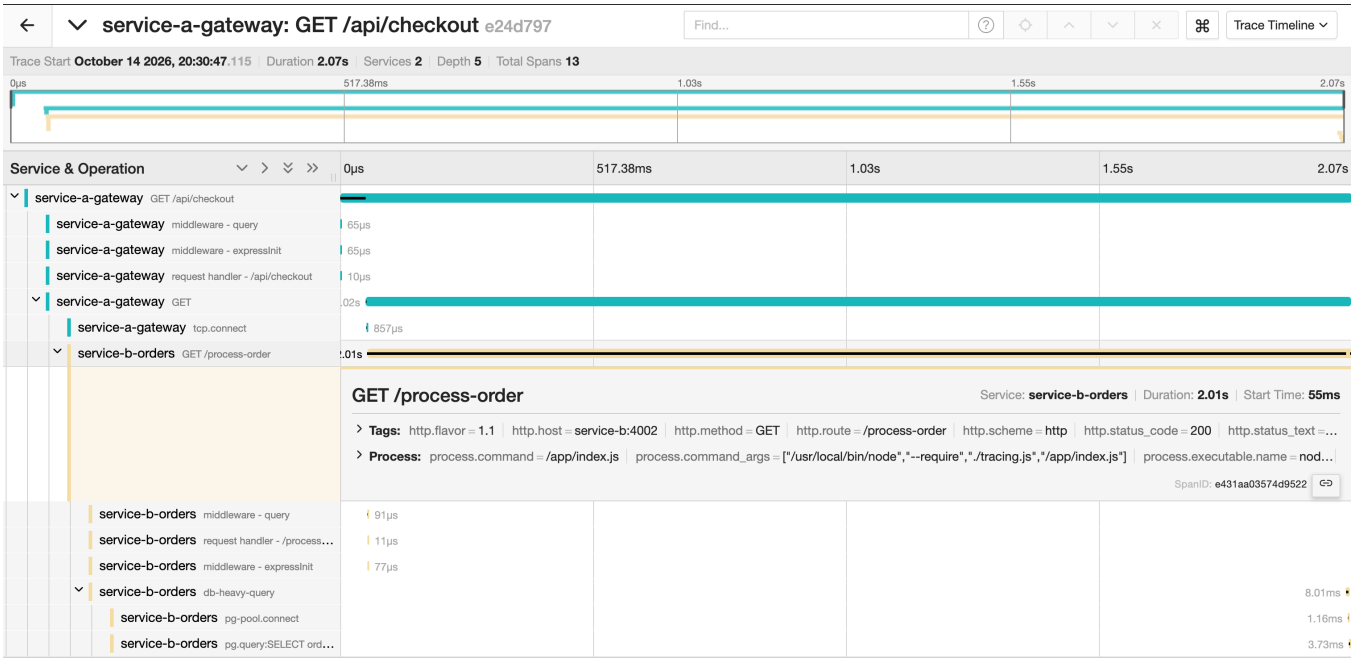


Рис. 3. Трасса `e24d797`. Длинный спан — `GET /process-order` сервиса `service-b-orders` (2,01 с).

Вопросы

- На каком сервисе и компоненте задержка? **`service-b-orders`**, обработчик **`GET /process-order`** (2,01 с).
- Причина — медленный SQL или код Node.js? **Код Node.js**: `setTimeout` (`delay = 2000`). `pg.query` в конце таймлайна занял 3,73 мс, между 56 мс и 2058 мс дочерних спанов нет.

2.3. Ошибка 500

Трасса `f49cd4a`, 2,06 с, HTTP 500, 10 спанов.

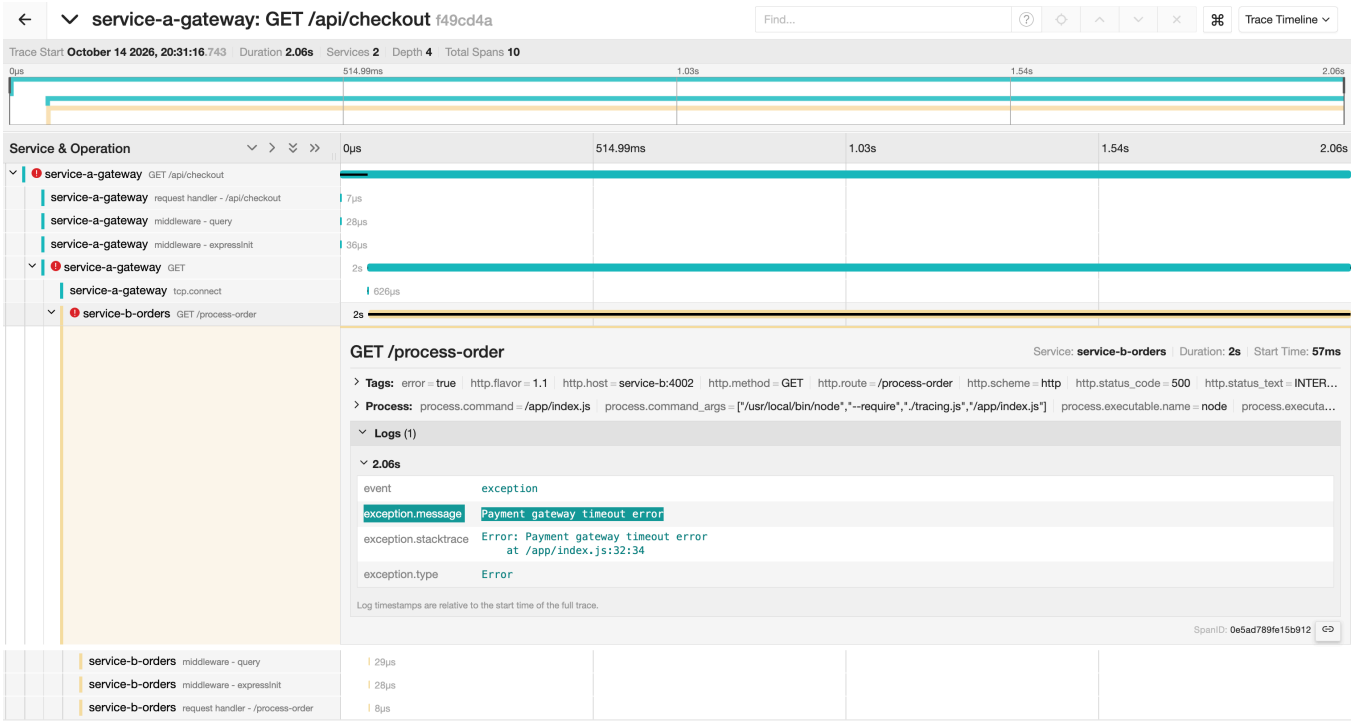


Рис. 4. Трасса `f49cd4a`. Лог спана `GET /process-order`.

Вопросы

- Какой сервис сгенерировал ошибку? **`service-b-orders`** (`GET /process-order`, статус 500).

- Какой `exception.message` в логе спана? **`Payment gateway timeout error`** (`/app/index.js:32:34`).
- Дошёл ли запрос до PostgreSQL? Нет: спанов, связанных с БД нет

Исправление

В `service-b/index.js` поставлено `isError = false`.

После `docker-compose up -d --build service-b` все 10 запросов вернули `SUCCESS`. Ошибок 500 нет.

```
❯ for i in {1..10}; do
  echo -n "Request #$i: "
  curl -s http://localhost:4001/api/checkout
  echo ""
  sleep 0.5
done
Request #1: Service A Success -> {"status":"SUCCESS","delay_applied_ms":2000,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status"
:"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10
-14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #2: Service A Success -> {"status":"SUCCESS","delay_applied_ms":100,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status":
"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10-
14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #3: Service A Success -> {"status":"SUCCESS","delay_applied_ms":100,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status":
"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10-
14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #4: Service A Success -> {"status":"SUCCESS","delay_applied_ms":100,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status":
"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10-
14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #5: Service A Success -> {"status":"SUCCESS","delay_applied_ms":2000,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status"
:"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10
-14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #6: Service A Success -> {"status":"SUCCESS","delay_applied_ms":100,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status":
"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10-
14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #7: Service A Success -> {"status":"SUCCESS","delay_applied_ms":2000,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status"
:"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10
-14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #8: Service A Success -> {"status":"SUCCESS","delay_applied_ms":100,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status":
"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10-
14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #9: Service A Success -> {"status":"SUCCESS","delay_applied_ms":100,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status":
"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10-
14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
Request #10: Service A Success -> {"status":"SUCCESS","delay_applied_ms":100,"orders":[{"id":1,"item_name":"Cloud Server vCPU-2","status"
:"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"},{"id":2,"item_name":"Object Storage 100GB","status":"PENDING","created_at":"2026-10
-14T17:30:06.241Z"},{"id":3,"item_name":"Load Balancer Basic","status":"COMPLETED","created_at":"2026-10-14T17:30:06.241Z"}]}
```

Рис. 5. 10 запросов после пересборки `service-b`: все ответы `SUCCESS`.

Выводы

Задержка 2 с — `setTimeout` в `service-b-orders`, SQL занимает единицы миллисекунд. Ошибка 500 — тот же сервис, `Payment gateway timeout error`, до базы запрос не доходит. После пересборки `service-b` с `isError = false` серия из 10 запросов прошла без 500.